Mortar and Artillery Variants Classification by Exploiting Characteristics of the Acoustic Signature

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Abstract

Feature extraction methods based on the discrete wavelet transform and multiresolution analysis facilitate the development of a robust classification algorithm that reliably discriminates mortar and artillery variants via acoustic signals produced during the launch/impact events. Utilizing acoustic sensors to exploit the sound waveform generated from the blast for the identification of mortar and artillery variants. Distinct characteristics arise within the different mortar variants because varying HE mortar payloads and related charges emphasize concussive and shrapnel effects upon impact employing varying magnitude explosions. The different mortar variants are characterized by variations in the resulting waveform of the event. The waveform holds various harmonic properties distinct to a given mortar/artillery variant that through advanced signal processing techniques can employed to classify a given set. The DWT and other readily available signal processing techniques will be used to extract the predominant components of these characteristics from the acoustic signatures at ranges exceeding 2km. Exploiting these techniques will help develop a feature set highly independent of range, providing discrimination based on acoustic elements of the blast wave. Highly reliable discrimination will be achieved with a feed-forward neural network classifier trained on a feature space derived from the distribution of wavelet coefficients, frequency spectrum, and higher frequency details found within different levels of the multiresolution decomposition. The process that will be described herein extends current technologies, which emphasis multi modal sensor fusion suites to provide such situational awareness. A two fold problem of energy consumption and line of sight arise with the multi modal sensor suites. The process described within will exploit the acoustic properties of the event to provide variant classification as added situational awareness to the solider.

Keywords: Neural Network, Artillery, Impact, Launch, Statistical Analysis, Acoustic Sensors, and Artillery

1. INTRODUCTION

A network of acoustic sensors can be used to detect and localize artillery launch and/or impact events, and can provide a capability to classify the types of events that have occurred in a specified region provide both warning and identification of threats found in the area. The added information integrated into the common operating picture would assist field commanders to (more) accurately plan and execute measures that must be taken in order to ensure the proper actions for a mission. Such a warning system would also help reduce the time-consuming, manpower intensive and dangerous tasks associated with identifying the events, allowing a field commander to make rapid and accurate judgments that insure greater safety and minimize danger for the soldiers.

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The reliability associated with using acoustic sensor technologies to discriminate between artillery variants/types using features that remain robust over long-range wave propagation, firing time, and detonation point (air/ground) are developed and discussed herein. The results described are based on an analysis conducted using acoustic signature data collected during a data collection exercise in the Spring of 2005. The artillery blast data was collected at YUMA Proving Ground (YPG) on 07-11 March 2005 in a field test experiment conducted by the Acoustic Center of Excellence (ACOE) from ARDEC (Picatinny Arsenal). The purpose of the experiment was to gather artillery and mortar data and subsequently investigate whether launch events could be distinguished from impact events using acoustic sensor technologies. The field test included the launch and detonation of several variants of artillery, comprising of various rounds and included mortars; High Explosive (HE) variants. The data set comprised of 52 IM event signatures and 125 LA event signatures collected using microphones emplaced at two sensors sites, S1 and S2, shown in Figure 1. S1 was located approximately 300m from the firing position, while S2 was emplaced 1000m from the firing position and 700m from the impact zone.

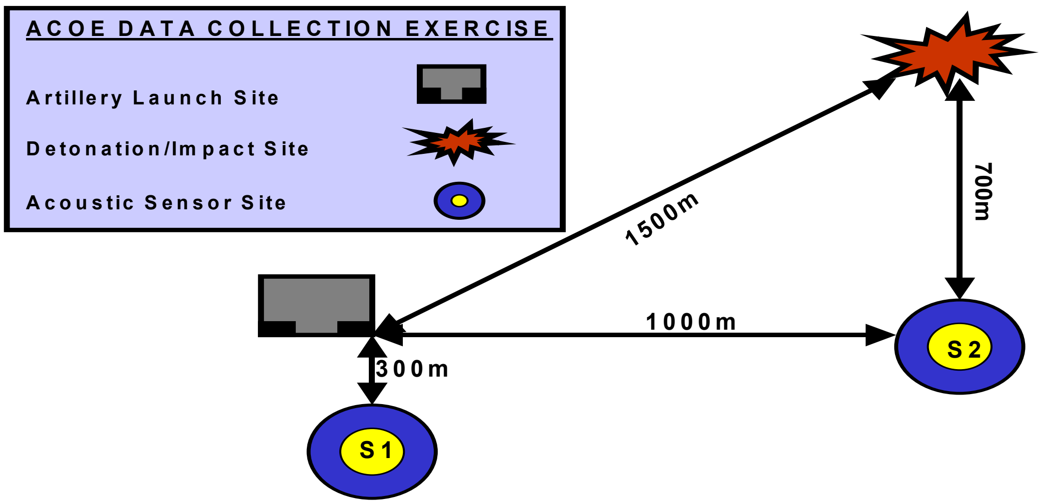

Figure 1 –Layout of Sensor Emplacement at Yuma Proving Grounds for Data Collection 1.

A second data collection exercise was conducted in January 2006 at Yuma Proving Grounds. The purpose of this data collection was to enhance the acoustic signature library. The acoustic recordings collected added a new class of mortars to the data base, primarily launches. The data collection also allowed us to extend the ranges of the acoustic microphone to provide greater standoff distance for classification. Four acoustic sensors were emplaced by the ACOE from ARDEC at sensor sites S1-S4 with ranges varying from 750m to 2000m from the firing position, shown in Figure 3.

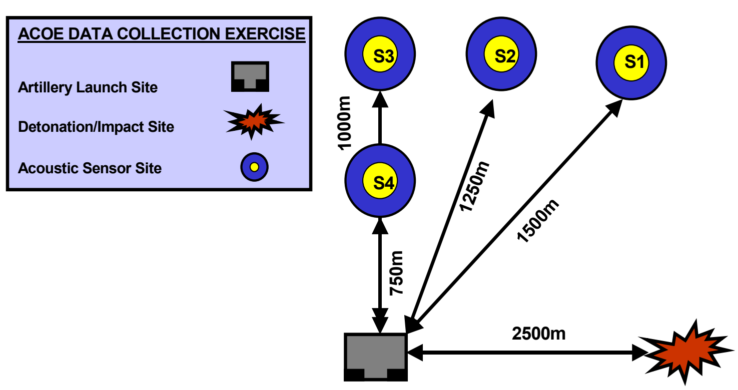

Figure 2 –Layout of Sensor Emplacement at Yuma Proving Grounds for Data Collection 2.

A third data collection exercise was conducted between May 31st and June 1st 2006 at Yuma Proving Grounds. The purpose of this data collection was two fold, first to increase the number of acoustic signatures in our library to continue to build our database. The second portion of the experiment was to conduct additional analysis of the influence of range variance on sound propagation from the acoustic source and the affects on the acoustic characteristics to develop a more robust feature space. Over the two-day data collection the assorted lots of mortar rounds were fired at various distances, ranging from 1000m-7500m. The air temperature was significantly higher from previous data collections, (100^0-106^0 Fahrenheit). Testing occurred under clear skies from the period of 0900-1500hrs. Figures 3 and 4 show the sensor deployment schemes for the two-day data collection are shown with approximate distances for the three acoustic sensors that were emplaced by the ACOE from ARDEC; S1, S2, and S3. The distances from the firing point and impact zones to the sensor sites are shown in figures 3 and 4.

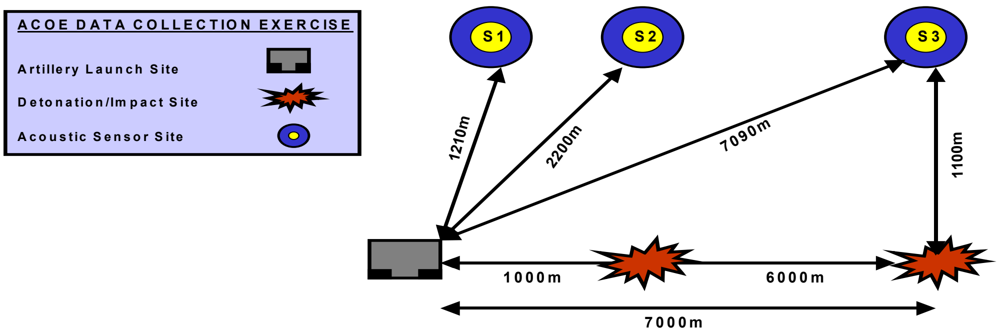

Figure 3 – Layout of Sensor Emplacement at Yuma Proving Grounds May 31st, 2006 for Data Collection 3.

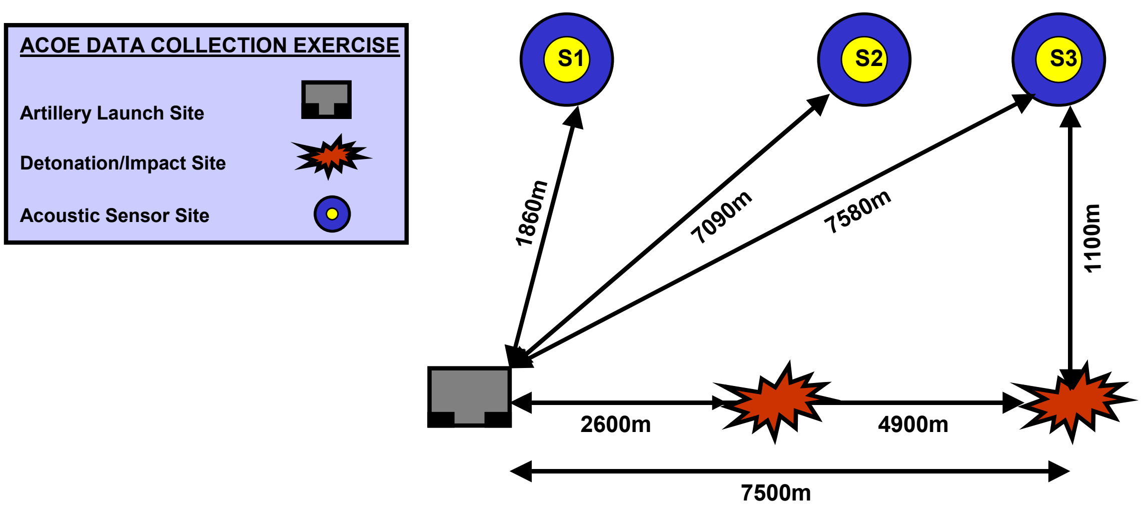

Figure 4 – Layout of Sensor Emplacement at Yuma Proving Grounds June 1st, 2006 for Data Collection 3.

Figure 5 shows three typical acoustic signatures for launches and impacts from various ranges to the sensors. The most notable attributes within each type of blast are readily identified. High frequency precursors that occur prior to the main blast of an IM type event are shown in Figure 5b. These precursors are attributed to various explosive and concussive properties within the blast, which degrade with long-range propagation. Note that the high frequency precursors are absent prior to the main blast as shown in

Figure 5. In this work, we identify range invariant features found within the multiresolution decomposition of the corresponding signatures in an attempt to further promote disparate sensor technology for artillery event discrimination. The subsequent development of a feature space representation for classification is based upon data obtained during the three series of data collections at YPG We show that details extracted via different levels of decomposition produce repeatable properties that can be processed and combined to ensure reliable discrimination between these events with various caliber of artillery at varying ranges.

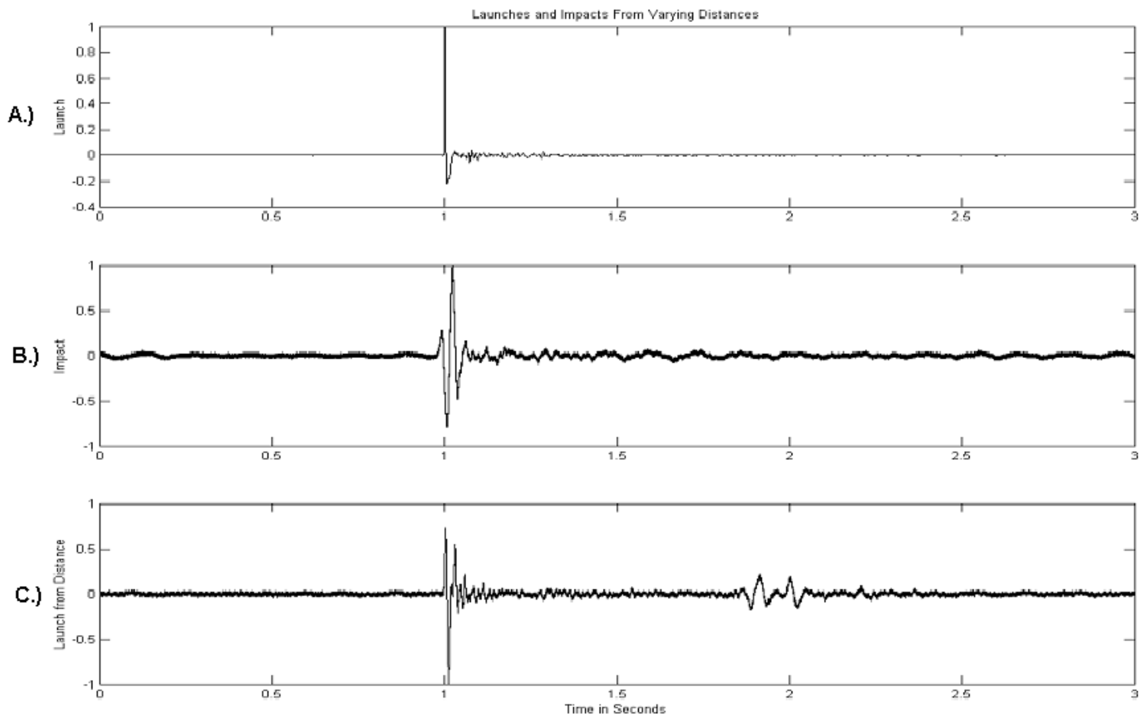

Figure 5 – Acoustic Signatures from YPG (a) Mortar launch recorded at S1, (b) Mortar impact recorded from S2, (c) Mortar launch recorded at S2.

2. DATA PROCESSING

Statistical analytical methods are used as a basis to identify distinct, disjoint feature sets that remain consistent for a given class of round, and do not degrade dramatically with long-range propagation. The complex, non-stationary response of the signatures to be analyzed, which are categorically poor candidates for feature extraction and segmentation via Fourier analysis or the short-time Fourier Transform. On the other hand, the non-stationary, transient and often oscillatory nature of these signals is well represented with a temporal representation identifying the acoustic baseband signature and developing a series of non-amplitude characteristics describing a given acoustic blast wave. Exploiting the transient nature of the source surface wave providing a scalable time representation, we process the raw data into manageable sizes.

After an impulsive event is detected, a three-second snippet is taken around the maximum. Various preprocessing steps are performed on the data. The first step is to remove any DC offset. The recorded acoustic signatures are down sampled. For training purposes it was convenient to convert the 32768 samples/sec to 8192 samples/sec. Although there was a small loss in information (particularly in the rapid onset as the acoustic wave first passes over the microphone) it was deemed that the benefits of working with less data outweighed any corresponding loss of information. The data was also normalization by scaling the recording to make the largest peak equal to unity.

The original data cuts were obtained by first estimating a noise floor and accepting a recording when the recording exceeded some dynamic threshold. The recording is taken 1 second from the maximum peak in the characteristic N-wave and 2 seconds forward from the same point providing the data snipet. Many standard approaches (e.g. the median absolute difference technique) are well know in the art and can be used for impulse detection. After a recording has been cut it is desirable to find a consistent reference point. Explosions generally have a characteristic N wave shape consisting of the following; a rapid onset followed by a rapid decrease of pressure until reaching a negative peak then slowly increasing back to zero.

2.1 DERIVING DISTINCTIVE ACOUSTIC CHARACTERISTIC'S

Utilizing statistical analytical properties a series of distinctive characteristic elements extracted from acoustic signatures developing a series of statistical methods that provide representation of the various phenomena occurring in the temporal domain. These derived features describe the various characteristics of surface wave generated by the explosive blast event at the launch of an artillery or mortar shell.

The acoustic signature generated by the expulsion of an artillery/mortar round out of a gun tube during launch divides into a series of temporal based events. The natural sequence of a round being launched is acoustically described by the representative signature recorded by the microphone. The signature divided into a series of events based on the characteristic N-wave generated by the charge used to launch the artillery/mortar round. The blast envelope is broken into a series of lobes describing the dramatic change in pressure as the explosion concludes with some minimal burn at the tail end as the signature fades out.

The signature is broken into the following series of intervals using the N-shaped envelope as represented as p[n] where n is the number of samples from the reference point defined in the previous section (first zero crossing after the initial peak pressure), and that zero crossing is noted. The lobes of the blast are defined by the time intervals $[\tau_1\ \tau_2]$ and $[\tau_3\ \tau_4]$. The reference point was chosen to be the first zero crossing after the initial peak pressure, essentially the middle leg of the characteristic N-wave. The ratio of the energy in the interval $[\tau_1, \tau_2]$ to the energy in the interval $[\tau_3, \tau_4]$ generates three key features based on the equation defined by eq.1.

(1)

$$f_{13} \equiv \frac{\sum_{n=\tau_1}^{\tau_2} p[n]^2}{\sum_{n=\tau_3}^{\tau_4} p[n]^2}$$

Pressure squared roll off. For a given interval $[\tau_1, \tau_2]$ and constant α $(0 < \alpha < 1)$ this feature is defined as follows: The time index τ that makes the energy of p in the interval $[\tau_1, \tau_1+\tau]$ equal to α times the energy in the interval $[\tau_1, \tau_2]$ is taken as the feature
$f_{14} = \Delta\cdot\tau$. Δ is also defined to be 1/(sampling rate) and is in several expressions to make the actual sampling rate used less relevant defined by eq.2 and eq.3.

(2)

$$f_{14} = \Delta \cdot \tau$$

For τ satisfying:

(3)

$$\sum_{n=\tau_1}^{\tau_1+\tau} p[n]^2 = \alpha \sum_{n=\tau_1}^{\tau_2} p[n]^2$$

The slope of a least squares linear fit to pressure over the interval $[\tau_1, \tau_2]$:

(4) $$f_{15} \equiv \frac{lS_{np} - S_n S_p}{lS_{nn} - S_n{}^2}$$

Each of the elements is defined in equations 5-9 over the defined time interval with various points on the characteristic N-wave.

(5) $$l = \tau_2 - \tau_1 + 1,$$

(6) $$S_n = \sum_{n=\tau_1}^{\tau_2} (\Delta n),$$

(7) $$S_{nn} = \sum_{n=\tau_1}^{\tau_2} (\Delta n)^2,$$

(8) $$S_p = \sum_{n=\tau_1}^{\tau_2} p[n],$$

(9) $$S_{np} = \sum_{n=\tau_1}^{\tau_2} \Delta \cdot n \cdot p[n]$$

The centroid of pressure squared over the time interval $[\tau_1, \tau_2]$:

(10) $$f_{21} \equiv C \equiv \frac{\sum_{n=\tau_1}^{\tau_2} \Delta (n - \tau_1) \cdot p[n]^2}{\sum_{n=\tau_1}^{\tau_2} p[n]^2}$$

The feature space proposed is comprised of primitives derived from the normalized energy distributions within the normalized original signature, centered about the maximum value of the blast wave that results in various power distributions characteristic to the event. Feature extraction methods yield differences that become visible with the varying energy distribution found within the events, including the energy distributions over the given period while remaining amplitude independent.

3. FEATURE OPTIMIZATION

It was desired to determine the optimal set of features and corresponding parameters. It is desired to obtain a small number of features that yields the most amount of information. A brute force search quickly becomes computationally prohibitive as the number of features increases. The approach taken was to define a measure of fitness and search over all the features and parameters for a single feature. The feature and corresponding parameters that yields the greatest measure of fitness is then recorded. Subsequent features are then added to the existing set, and the merit of each combination is evaluated, and new feature/parameters are selected in turn. We do not forbid the possibility that the same feature with a different data cut is best (i.e. features can occur multiple times). This process is repeated until the desired number of features is reached (in our case 6 features were selected).

In order to evaluate the usefulness of each set of features a metric is needed. The metric chosen was the Mahalanobis distance. For the two class case with means, μ_1 and μ_2 and pooled covariance Σ given by:

(11)
$$\Sigma \equiv \frac{(N_{C1}-1)\Sigma_{C1}+(N_{C2}-1)\Sigma_{C2}}{N_{C1}+N_{C2}-2}$$

The Mahalanobis distance is defined as:

(12)
$$d(c1,c2) \equiv \sqrt{(\mu_1-\mu_2)^T \cdot \Sigma^{-1} \cdot (\mu_1-\mu_2)}$$

For each possible set of features and parameters, the Mahalanobis distance between each combination of classes is computed. The minimum of these Mahalanobis distances are taken as a measure of fitness for a given set of feature/parameters to then proceed to the next step of classification.

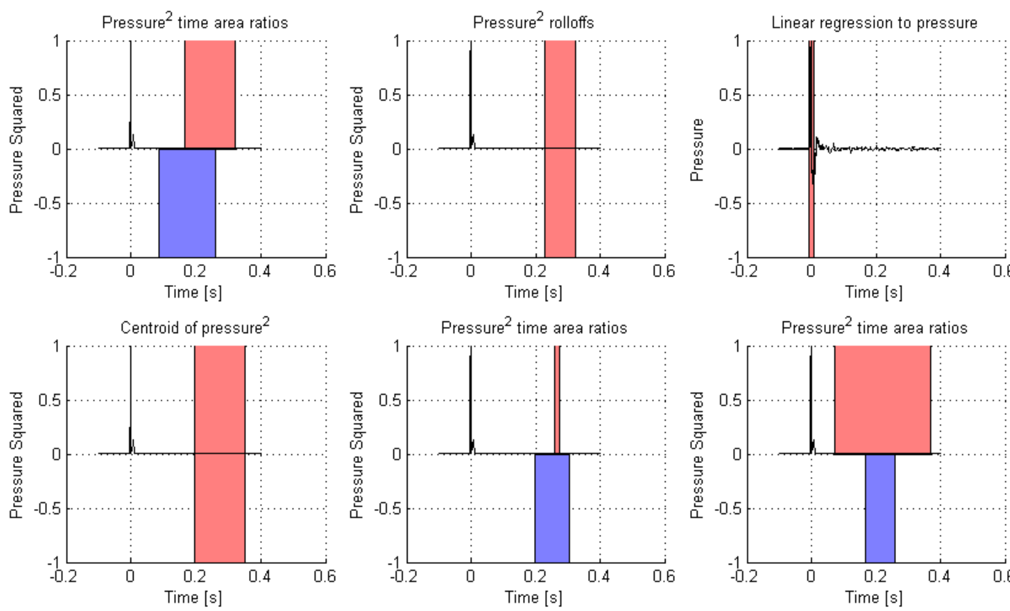

Figure 6 – A representation of the derived, defined feature space across an acoustic signature of high-explosive event..

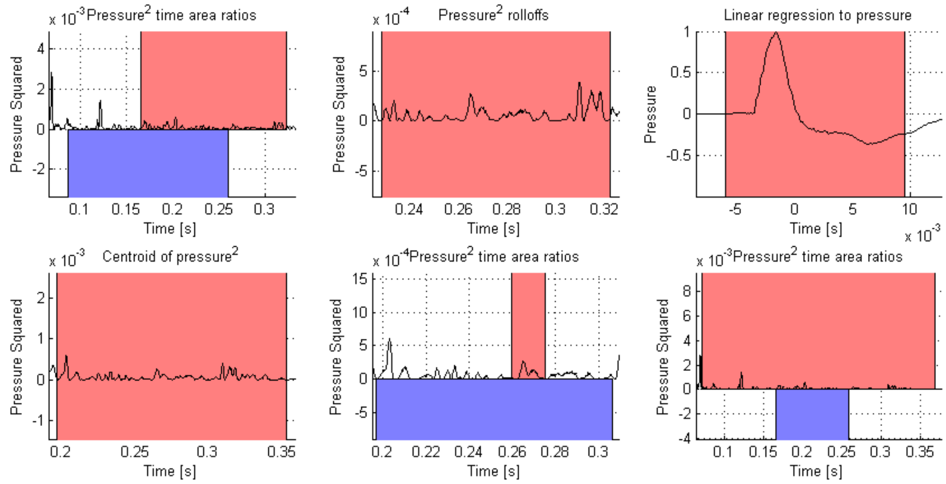

Figure 7 – A zoomed representation of the derived, defined feature space providing greater detail.

4. NEURAL NETWORK CLASSIFIER

Neural networks have become a powerful tool for solving difficult classification (mapping) problems with a proven ability to realize non-linear discriminant functions and complex decision regions that are often required to ensure separability between classes. The use of neural networks for classification is well documented and convergence requirements for training are well known. The standard multilayer feedforward neural network shown in Figure 8 was chosen due to its ability to learn mappings of any complexity, provided that the number of hidden layer neurons is sufficient to accommodate the number of separable regions that are required to solve a particular classification problem. In general, the network contains N_i inputs, N_h hidden layer neurons and N_O output layer neurons with no interconnections within a single layer. In the three-layer network shown, the connection weight between the i^{th} input and j^{th} hidden layer neuron will be denoted as w_{ij} and v_{ij} will be used to denote the connection weight between the j^{th} neuron in the hidden layer and k^{th} output layer neuron. A bias for the j^{th} hidden layer neuron is denoted as b_j and the bias for the k^{th} output neuron is $\hat{b}_k$. x^p denotes the set of P, N-tuple feature vectors $x_n \in \Re^N$, $x^p = \lfloor x_1^p, x_2^p, \ldots, x_N^p \rfloor$ used to discriminate between the different classes of artillery.

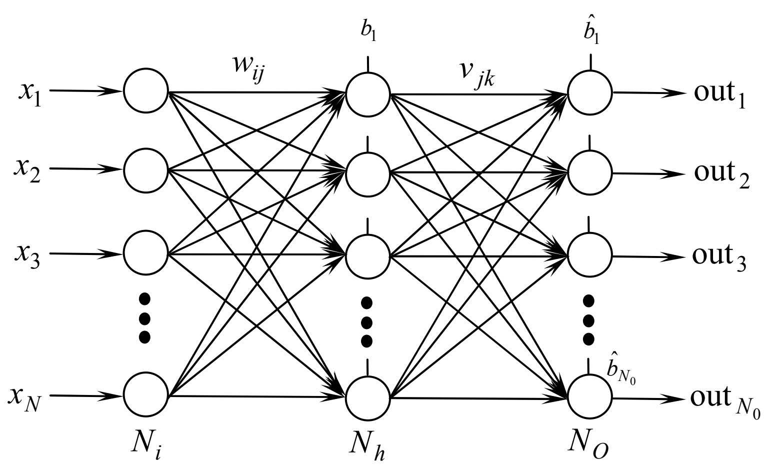

Figure 8– Standard Multilayer Feedforward Neural Network.

The input layer units or “neurons” propagate signals to the hidden layer but do not perform any computations. Neurons in the hidden layer and the output layer compute their response by taking a weighted sum signals from the previous layer plus a bias, and then passing the sum through an activation function. For example, the output of the j^{th} neuron in the hidden layer upon the presentation of the p^{th} input pattern x^p is given by $\text{out}_j^P = f\left(\text{net}_j^P\right)$ where $\text{net}_j^P = \sum_{i=1}^{N} w_{ij} x_i^P + b_j$, and f is an activation function, that was chosen to be the sigmoid function

$$f(x) = \frac{2}{1+\mathrm{e}^{-\mathrm{x}}} - 1 \qquad (13)$$

The neural network classifier used to discriminate between variant artillery and mortar round types. The neural network was trained using the generalized delta rule or back propagation algorithm. The algorithm sequentially adjusts the interconnection weights within the network, subsequent to the application of all patterns in a training set, a routine commonly referred to as an epoch. In general, when an input pattern x^p from a training set is presented to the network, it produces and output that is different from the target value, say d^P. The error for the specific pattern is defined as the squared error $E_P = \frac{1}{2}\sum_{k=1}^{N_0}\left(d_k^P - \text{out}_k^P\right)^2$. An unconstrained nonlinear optimization performed to minimize the total error function of the network

$$E(w) = \sum_{k=1}^{P} E_P(w) \qquad (14)$$

through the incremental computation of the gradient of the error in equation (14), and successive adjustment of the interconnection weights so as to achieve the global minimum error corresponding to $E(w) = 0$. In this implementation the Levenberg-Marquardt algorithm was used. To train the network the error function is defined as the sum of error functions defined above for all input/output pairs of the training set. Levenberg-Marquardt is hybrid technique that utilizes a parameter that changes during training to vary between steps defined by the Gauss-Newton algorithm and standard gradient descent.

5. RESULTS

EXPERIMENT 1 RESULTS

Features were extracted from the data sets using standard statistical analysis tools for signal processing as described in sections two and three, and the 6-tuple feature vector $\zeta^p = \left| f_1, f_1, f_1, f_4, f_5, f_6, \right|$ was constructed according to equations (1), (3), (4), and (10), respectively. Optimization techniques were utilized to exploit the most robust and ideal set of features to provide maximum seperability, giving the most amount of information. The optimization method defined a measure of fitness for each of Experiments were then conducted to measure the separability of the feature space and benchmark the performance of the proposed neural network classifier. The architecture of the neural network consisted of five hidden layer neurons trained using 2/3 of randomly selected vectors from a total of 423 signatures made available from the multiplicity of YPG exercise between March of 2005, YPG 1 and June of 2006, YPG 3 . The neural network has six input nodes and five output nodes to separate the various rounds based on caliber. The training set comprised of 2/3 of the total data set composed of various mortars and artillery caliber rounds at the diverse sensor sites. When tested against the remaining 1/3 total data signatures composed of artillery and mortars , the networked correctly classified 132 out of 141 rounds with the correct caliber based on the launch events. The overall performance of network was 93.62% for correct artillery and mortar variant classification; each individual round had the following classification rates, 86% for Type A, 96% for Type B, 95% for Type C, 100% for Type D, and 77% for Type E rounds. The events were classified as a certain round caliber when the output from the sigmoid activation function in equation (14) was greater than 0 and less then 0 for the series of outputs giving an expression in 1 and -1 values.

	Classified Type A	Classified Type B	Classified Type C	Classified Type D	Classified Type E	Correctly Classified
TypeA	6	0	1	0	0	85.7%
TypeB	0	45	1	0	1	95.7%
TypeC		1	60	2		95.2%
TypeD	0	0	0	11	0	100%
TypeB	0	0	0	3	10	77%

Figure 9– Classification Results Table to Discriminate Various Round Calibers.

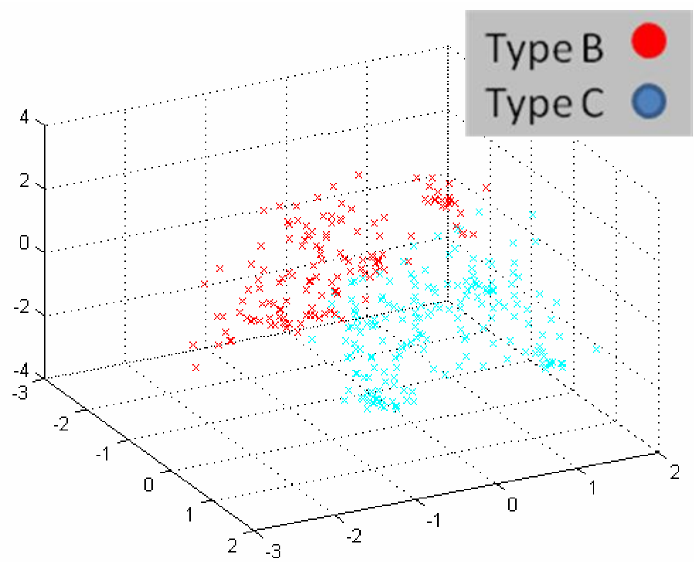

Figure 10– A 3-tuple Representations of Seperability of Artillery/Mortar Based on Caliber Type.

6. CONCLUSIONS

In this paper, feature extraction methods based on the statically analysis methods for acoustic signals and temporal space analysis facilitate the development of a robust classification algorithm that affords reliable discrimination between various calibers of artillery and mortar round types based on launch event acoustic signatures. The acoustic signature generated by the various charges utilized to expel a round out of a firing tube affords a series of characteristic features amplitude independent, and range independent that can discriminate various caliber rounds. Previous methods for artillery/mortar classification based on launch events were based on visual cues rather than remote acoustic methods, which vary on proximity to the event and line of sight. Amplitude dependant features, which vary dramatically due to signal attenuation and distortion resulting from various propagation effects, are minimized through the use of signal processing techniques that exploit class specific attributes that remain constant within the signature data and furthered through statistical analysis [4].

The non-stationary, transient and often oscillatory nature of artillery blast signals may be represented efficiently with a temporal base that effectively capture the time-frequency distribution of such signal components. Classification results using a neural network trained on the feature set show reliable discrimination of artillery and mortar events from various distances is 94% based singularly on launch events for events between 700 - 7500 meters.

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